

Evidence for Intermediate-Mass Black Holes From Microlensing Signatures in CHIME/FRB Catalog 2

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ABSTRACT

Intermediate-mass black holes (IMBHs) are the missing link in the cosmic hierarchy of black holes, bridging the gap between stellar-mass black holes and supermassive ones. They also serve as unique laboratories for testing strong-field gravity and are prime targets for future multi-messenger observations. However, IMBHs are a population that has remained notoriously difficult to detect. The microlensing effect of fast radio bursts (FRBs) can serve as a clean and powerful method to probe IMBHs. In this work, we develop a pipeline to search for microlensed FRBs based on their dynamic spectra and apply it to the CHIME/FRB Catalog 2. Two microlensing signatures have been identified in two separate sources, i.e. FRB 20190131D and FRB 20211115A. The inferred lens masses for these two signatures are $\sim [280 - 467] M_{\odot}$ and $\sim [539 - 609] M_{\odot}$, respectively. Here we interpret them as evidence for IMBHs. If there are no intervening structures—such as galaxies or clusters—along the line of sight for these two sources, the two identified IMBHs might be isolated and of primordial origins. In that case, we obtain primordial black holes (PBHs) within these two mass ranges would constitute $\sim 4\%$ of dark matter. Moreover, if these two candidates are not genuine lensing signatures, the abundance of intermediate-mass PBHs with masses $> 300 M_{\odot}$ is constrained to be $\sim 13\%$ at 95% confidence level. Therefore, more comprehensive observational information for FRBs, together with a deeper understanding of whether the intrinsic emission mechanisms of FRBs can produce lensing-like signals, will be crucial for establishing this effect as a powerful tool for probing (primordial) IMBHs.

Keywords: Fast radio bursts, Microlensing, Primordial black holes

1. INTRODUCTION

Intermediate-mass black holes (IMBHs), defined as black holes with masses between 10^2 and $10^5 M_{\odot}$, occupy a critical yet poorly explored regime in the cosmic black hole mass function, bridging the gap between stellar-mass black holes ($\lesssim 10^2 M_{\odot}$) and supermassive black holes ($\gtrsim 10^6 M_{\odot}$). Candidate IMBHs have been reported through multiple observational channels, each with distinct strengths and limitations: (i) radiative accretion signatures from ultraluminous X-ray sources (Tremou et al. 2018; Lin et al. 2018), which can also be explained by super-Eddington accretion onto stellar-mass black holes; (ii) stellar kinematics from integral field spectroscopy (Pechetti

et al. 2022; Häberle et al. 2024; Huang et al. 2025), limited by spatial resolution requirements; (iii) gravitational-wave detections of binary black hole mergers in the IMBH regime, most notably GW190521 and GW231123, which provide the most direct evidence by encoding component masses without astrophysical modeling degeneracies (Abbott et al. 2020; Abac et al. 2025a); and (iv) microlensing of extragalactic transients (Paynter et al. 2021; Abac et al. 2025b; Xiong et al. 2026), whose interpretation depends on the assumed lens model. Despite these efforts, robust confirmation of a substantial IMBH population remains challenging, as most candidates suffer from degeneracies with other astrophysical systems, and the IMBH mass function remains observationally unconstrained.

From a theoretical perspective, one of the leading formation channels for IMBHs is dynamical assembly in dense star clusters, which circumvents the mass limits of stellar collapse. Three key mechanisms have been identified: (i)

runaway stellar collisions in dense cluster cores producing a very massive star ($\gtrsim 10^3 M_\odot$) that collapses directly into an IMBH (Portegies Zwart et al. 2004), most efficient at low metallicities ($Z \lesssim 0.1 Z_\odot$) and high cluster densities ($\gtrsim 10^5 M_\odot \text{pc}^{-3}$); (ii) hierarchical mergers of stellar-mass black holes that gradually build up an IMBH (Arca Sedda et al. 2023; Paiella et al. 2026), with efficiency depending on cluster retention fraction and binary availability; and (iii) tidal disruption events that contribute to black hole growth over time (Rizzuto et al. 2023), though likely subdominant. The relative importance of these channels remains an active area of investigation, with implications for the predicted IMBH mass function and its redshift evolution.

In addition to the above-mentioned formation channels, primordial black holes (PBHs), formed in the early Universe via the collapse of primordial density fluctuations (Hawking 1971; Carr & Hawking 1974; Carr 1975), are also a possible origin of IMBHs. Meanwhile, PBHs are a well-motivated dark matter candidate whose mass range spans over 50 orders of magnitude—from the Planck scale ($\sim 10^{-5} \text{g}$) to supermassive black hole masses ($\gtrsim 10^9 M_\odot$). Recent multi-wavelength and multi-messenger advances have positioned PBHs as a compelling theoretical alternative for interpreting various astrophysical phenomena, including the origin of IMBHs. In GW astronomy, the LIGO-Virgo-KAGRA Collaboration has detected several BBH merger events with component masses in the IMBH regime (Abbott et al. 2020; Abac et al. 2025a), most notably GW190521 and GW231123, whose masses and spins are difficult to explain via conventional stellar evolution but can be reproduced by the PBH merger channel when cosmological accretion is taken into account (De Luca et al. 2026b). Similarly, recent JWST observations have revealed a population of quasars at $z \gtrsim 6$ (e.g., Maiolino et al. 2024; Bogdan et al. 2024; Natarajan et al. 2024), with Abell2744-QSO1—hosting an SMBH of $\sim 5 \times 10^7 M_\odot$ with an extreme black hole-to-stellar mass ratio and low metallicity—posing severe challenges to conventional SMBH formation models (Maiolino et al. 2025; Juodbalis et al. 2025). These anomalies motivate alternative seeding scenarios, including PBHs, which could potentially account for such systems under certain accretion and evolution conditions (Dayal & maiolino 2025; Zhang et al. 2025; De Luca et al. 2026a). Extensive observational searches have placed constraints on the PBH abundance $f_{\text{PBH}} \equiv \Omega_{\text{PBH}}/\Omega_{\text{DM}}$ across a wide range of mass windows (Sasaki et al. 2018; Green & Kavanagh 2021; Carr et al. 2021; Carr & Kuhnel 2022; Carr et al. 2026); for masses $\gtrsim 100 M_\odot$, current limits from microlensing and GW observations constrain $f_{\text{PBH}} \lesssim 0.1$ –1, depending on the mass range and assumed mass function. Among various meth-

ods, gravitational lensing provides one of the most direct probes of PBH abundance, sensitive to masses from planetary scales ($\sim 10^{-7} M_\odot$) to SMBHs, and can be categorized into four types: (1) microlensing-induced flux variations of stars (Niikura et al. 2019; Mróz et al. 2024); (2) time-delayed multi-peak structures in transients such as FRBs and GRBs (Muñoz et al. 2016; Liao et al. 2020); (3) waveform distortions in coherent transients due to lensing interference (Jung & Shin 2019; Kader et al. 2022); and (4) angularly separated multiple images of persistent sources (Wilkinson et al. 2001; Zhou et al. 2022b).

Fast radio bursts (FRBs) are millisecond-duration radio pulses of cosmological origin, with isotropic-equivalent energies spanning $\sim 10^{36}$ – 10^{41} erg (Lorimer et al. 2007). Only $\sim 2\%$ – 3% repeat (Andersen et al. 2023; Abbott et al. 2026), likely a lower bound; the intrinsic fraction may exceed 50% (Yamasaki et al. 2023). The Galactic FRB associated with SGR 1935+2154 confirmed magnetar engines as one progenitor channel (Zhang 2020; Andersen et al. 2020; Bochenek et al. 2020; Lin et al. 2020), though the broader radiation mechanism remains debated. Despite these uncertainties, FRBs’ clean temporal structure, cosmological distances, and high event rate ($\sim 10^3$ – $10^4 \text{sky}^{-1} \text{day}^{-1}$ (Cordes & Chatterjee 2019; Petroff et al. 2019)) make them promising probes for cosmology (Deng & Zhang 2014; Liu et al. 2023; Wei et al. 2015; Wu et al. 2016; Li et al. 2018) and compact objects. Muñoz et al. (2016) proposed using FRB microlensing to probe IMBHs, and subsequent searches have been conducted (Liao et al. 2020; Zhou et al. 2022a; Krochek & Kovetz 2022; Chang et al. 2025; Xiong et al. 2026), identifying FRB 20190308C and FRB 20190320B as possible candidates—though both remain debated. As next-generation facilities (CHIME, DSA-2000, SKA) expand the FRB sample, FRBs will become increasingly powerful for constraining PBHs (Muñoz et al. 2016; Laha 2020; Oguri et al. 2023; Connor & Ravi 2023).

In this paper, we refine the microlensing search methodology and apply it to CHIME/FRB Catalog 2 (Abbott et al. 2026) to search for lensing signals and update their implications on IMBHs and abundance of PBHs.

2. FRB MICROLENSING CANDIDATES

In this section, we briefly introduce the current status of CHIME/FRB observations, present the method for searching and identifying microlensing signatures, and report the search results along with an analysis of the candidate properties.

2.1. FRB Observations in CHIME/FRB Catalog 2

The rapid increase in the number of verified FRBs is currently being driven by the operation of several wide-field radio telescopes, such as Canadian Hydrogen

Intensity Mapping Experiment (CHIME), Five-hundred-meter Aperture Spherical radio Telescope (FAST), the Australian Square Kilometre Array Pathfinder (ASKAP), and the Deep Synoptic Array (DSA). As a prominent example, the CHIME/FRB Collaboration initially released its first catalog containing 535 FRBs detected in less than one year (from July 25, 2018, to July 1, 2019) (Amiri et al. 2021). Subsequently, a second catalog (CHIME/FRB Catalog 2) was published, comprising 4539 FRBs observed with the CHIME telescope between July 25, 2018 and September 15, 2023 (Abbott et al. 2026). The second catalog includes all FRBs from the first catalog, with every event reprocessed using a uniform and improved analysis framework. The 4539 bursts originate from 3641 unique sources: 3558 non-repeating sources, 83 known repeating sources contributing a total of 981 bursts¹.

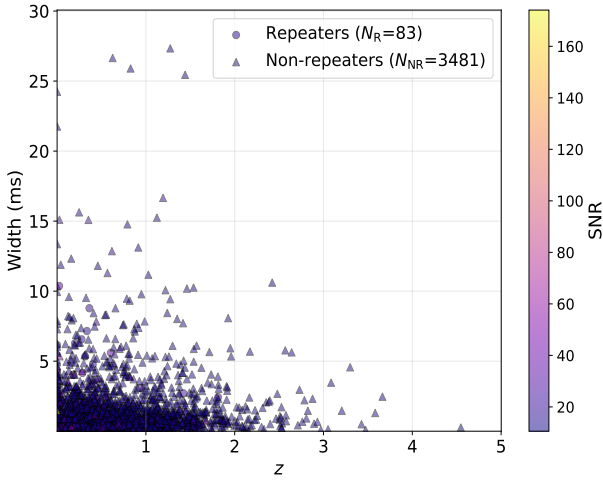


Figure 1. Two-dimensional distribution of inferred redshifts, width, and SNR for 3564 FRBs in CHIME/FRB catalog 2.

For a detected FRB, the dispersion measure (DM) is a key observational property. Theoretically, DM is the integral of the electron density along the radio pulse’s path. Observationally, it is derived from the arrival time difference between two photons at different frequencies. The high DMs of the first few poorly localized bursts suggested a cosmological origin for FRBs (Lorimer et al. 2007; Thornton et al. 2013). This was later confirmed when the repeating FRB 20121102A was localized to a nearby dwarf galaxy (Chatterjee et al. 2017; Tendulkar et al. 2017; Marcote et al. 2017). Therefore, the distance and the redshift can be roughly derived from the observed DM of a detected FRB, which is usually decomposed into four components: the Milky Way

(DM_{MW}), the intergalactic medium (DM_{IGM}), and the host galaxy including the source local environment (DM_{host} and DM_{src}), respectively

$$DM = DM_{MW} + DM_{IGM} + \frac{DM_{host} + DM_{src}}{1 + z}. \quad (1)$$

Here, we conservatively adopt a maximum value of $DM_{host} + DM_{src} = 200; pc, cm^{-3}$, which corresponds to the minimum inferred redshift for all host galaxies. The $DM_{IGM} - z$ relation, given by Deng & Zhang (2014) and approximated as $DM_{IGM} \approx 855z; pc, cm^{-3}$ (Zhang 2018), takes into account the helium ionization history and assumes an intergalactic medium (IGM) baryon fraction $f_{IGM} = 0.83$. This relation is statistically supported by the five localized FRBs available at the time (Li et al. 2020). The key observational features of FRBs and the inferred redshifts for the lensing scales, based on the currently public FRBs from the CHIME/FRB Catalog 2, are shown in Figure 1. It is worth noting that among the 3558 nonrepeating sources, 77 sources lack width and SNR information and are therefore excluded from further analysis owing to observational issues, such as the inability to fit burst morphology from heavy Radio Frequency Interference (RFI) or background confusion, events on lowsensitivity days (flagged by pulsar-based noise monitoring), and events during periods when the realtime L2/L3 classification was nonfunctional.

2.2. Searching for Microlensing Signatures

Previous studies have established the foundation for searching microlensed FRBs by exploiting characteristic signatures such as similar light-curve morphology and the absence of significant frequency drift (Liao et al. 2020; Zhou et al. 2022a), and have further applied correlation methods to search for such candidate events (Zhou et al. 2022a; Krochek & Kovetz 2022; Chang et al. 2025; Xiong et al. 2026). Building on these efforts, we develop a complete screening pipeline that systematically incorporates noise treatment and selection-bias mitigation, while accounting for the intrinsic features of microlensed FRBs. The pipeline consists of four sequential tests described in the following subsections: Autocorrelation Test, PeakSNR Test, FrequencyDrift Test, and Hardness Test. This approach is designed for the current CHIME/FRB Catalog 2 and can be readily extended to larger future FRB samples.

2.2.1. Autocorrelation Test

Among FRBs, the most likely lensing signals are those with clear multipeak structures (sub-burst); the CHIME/FRB Catalog 2 identified 340 FRBs exhibiting such morphologies (Abbott et al. 2026). For intrinsically short-duration FRBs (with a duration of 1 ms), a microlensed burst would appear as two distinct peaks. Their time delay can be clearly read from the light curve, as long as the

¹ The basic information for all FRBs in the CHIME/FRB Catalog 2 is available at <https://www.chime-frb.ca/catalog2>.

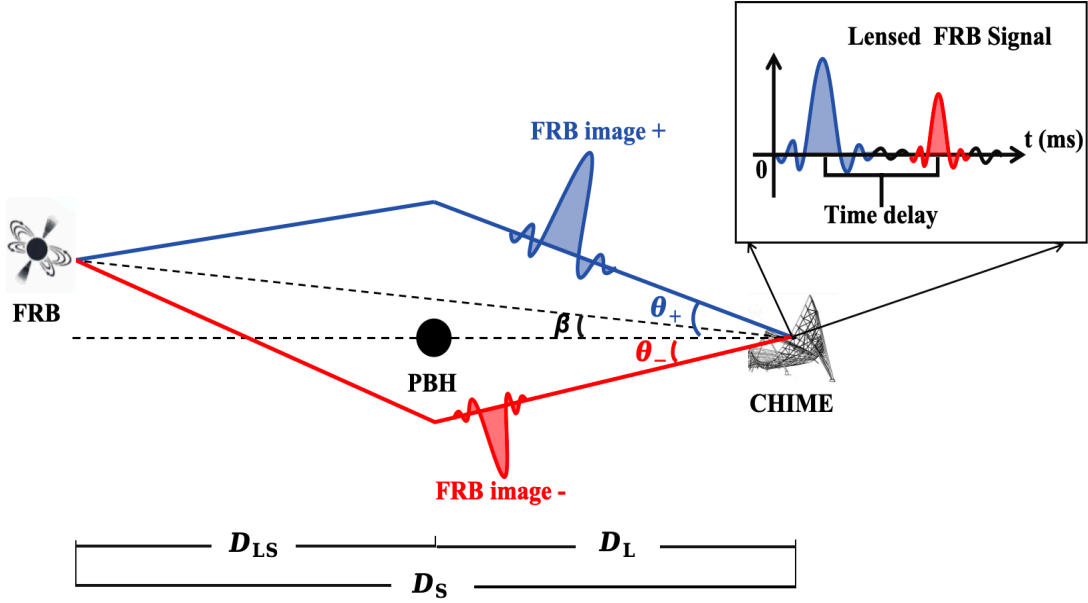


Figure 2. Schematic diagram of the FRB being lensed considering PBH as point-mass lens.

flux ratio is not too small. The light curve of a microlensed FRB will have an echo superimposed on it. To detect this from these FRBs with multi-peaks, we define the normalized auto-correlation function (ACF) of light curve

$$C(\delta t) = \sum_t \frac{\tilde{I}(t)\tilde{I}(t - \delta t)}{N_{\delta t}\sigma_I^2}, \quad (2)$$

where $\tilde{I}(t) = I(t) - \mu_I$ is intensities of the light curve, where σ_I is the standard deviation for the light curve $I(t)$, δt is relative displacement for autocorrelation, and $N_{\delta t}$ is the bin number of the light curve. The effectiveness of the autocorrelation analysis on lensed data can be shown as follows. Let $I_{\text{int}}(t)$ be the intrinsic light curve of the FRB event and $C(\delta t)$ its autocorrelation. In a specific point-mass lensing configuration as shown in Fig 2, we assign a time delay Δt and a flux ratio R_f to the second image, yielding a lensed light curve

$$I_{\text{lensed}}(t, \Delta t, R_f) \propto \frac{R_f}{R_f + 1} I_{\text{int}}(t + \Delta t) + \frac{1}{R_f + 1} I_{\text{int}}(t). \quad (3)$$

Using this formula together with Eq. (2), the ACF of the lensed signal exhibits spikes at $\delta t = -\Delta t, 0, +\Delta t$ with an amplitude ratio $R_f/(R_f^2 + 1) : 1 : R_f/(R_f^2 + 1)$. To distinguish lensing-induced spikes from noise, we define the sigma parameter for ACF of light curve (Ji et al. 2018; Paynter et al. 2021; Chang et al. 2025; Xiong et al. 2026)

$$\sigma_{\delta t} = \sqrt{\frac{1}{N_{\delta t}} \sum_{\delta t \in \Delta t} (C(\delta t) - G(\delta t))^2}, \quad (4)$$

where $C(\delta t)$ is the ACF to be examined, which may correspond to either a lensed or an unlensed signal, $G(\delta t)$

is Gaussian-smoothed version (standard deviation of the Gaussian kernel $\sigma = 3$) of $C(\delta t)$ ², and $\Delta t \in [0, N_{\delta t}\Delta t_{\text{min}}]$ (Δt_{min} represents the time resolution of CHIME, which is approximately 0.98 ms) denotes the time grid over which the calculation is performed. The parameter σ quantifies the overall spikiness of $C(\delta t)$ relative to the smooth template $G(\delta t)$. If we identify a δt for which $|C(\delta t) - G(\delta t)|$ exceeds a certain threshold i.e., 3σ , we can claim evidence for a spike induced by lensing. In Fig. 3 we present the results of this analysis for FRB 20181028A, revealing a spike at 15.68 ms that most likely corresponds to the time delay between the third and fourth peaks in the dynamic spectrum³.

2.2.2. Peak-SNR Test

In addition to the screening criteria described above, for point-mass lenses such as PBHs illustrated in Fig. 2, the lensed double peaks must satisfy the condition that the main peak arrives before the secondary peak. Therefore, we define the peak SNR of a peak in the dynamic spectrum

² We have verified that the Gaussian smoothing parameters used in the selection process have little effect on the resulting FRB candidates, and the two potential events (FRB 20190131D and FRB 20211115A) remain unaffected. While different fitting methods (e.g., SavitzkyGolay filtering of the ACF (Paynter et al. 2021; Chang et al. 2025; Xiong et al. 2026)) may introduce discrepancies, they do not exclude high-SNR lensed signals. Independent tests employing the SavitzkyGolay filter corroborate the persistence of above two candidates.

³ The dynamic spectra of all FRBs were obtained from the CHIME/FRB Catalog 2 Public Data, available at the Canadian Astronomy Data Centre: <https://www.canfar.net/storage/list/AstroDataCitationDOI/CISTL.CANFAR/25.0066/data>.

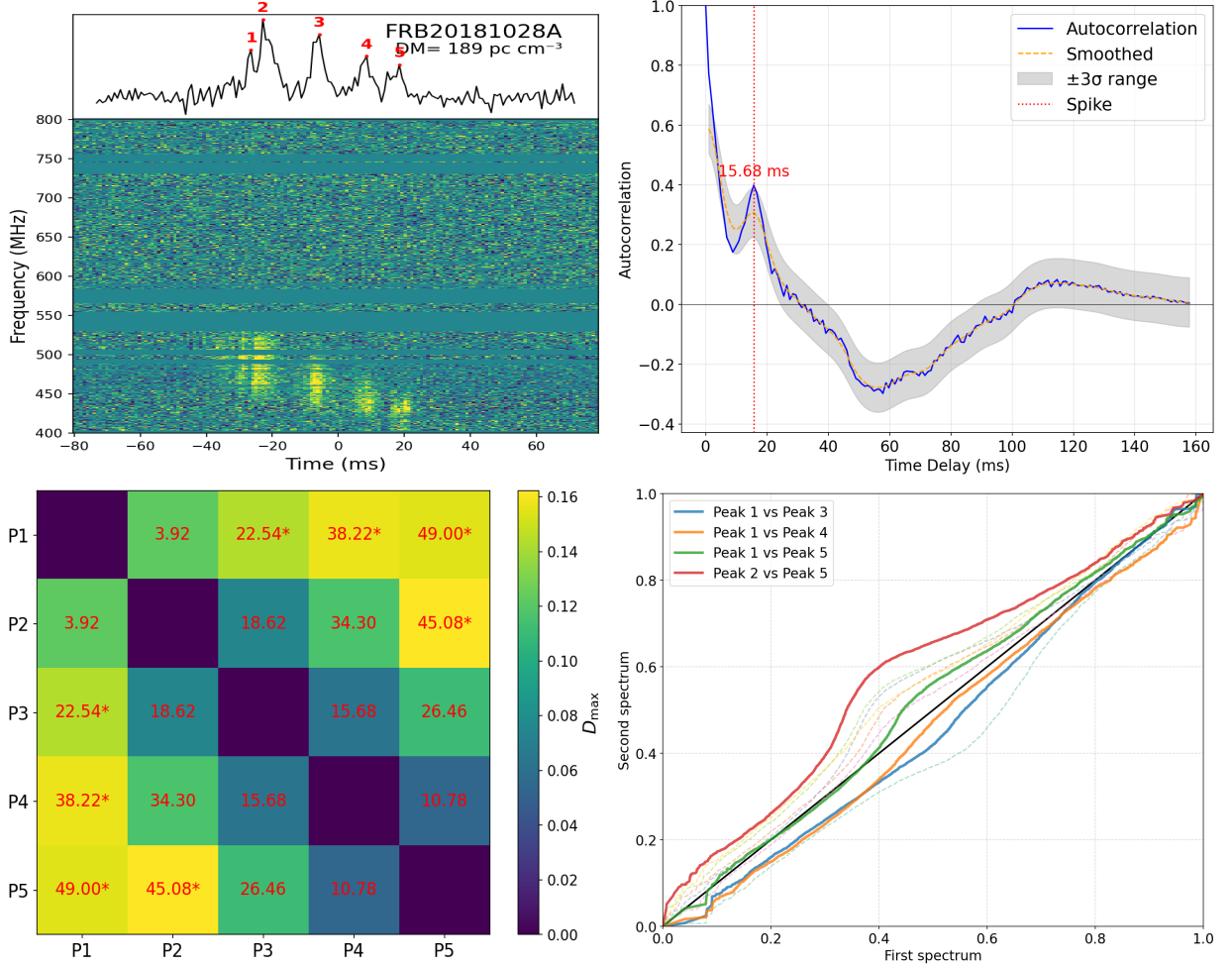


Figure 3. Upper Left: The dynamic spectrum of FRB 20181028A without the RFI. **Upper Right:** The blue line shows the autocorrelation of the light curve. The orange dashed line represents a Gaussian-smoothed version of the autocorrelation. The gray dash-dotted lines indicate the 3σ . The red dashed line marks a location with a strong autocorrelation exceeding the 3σ threshold, corresponding to a time delay of $\Delta t = 15.68$ ms. **Lower Left:** As indicated by the peaks marked in the dynamic spectrum, the heatmap shows the differences in the CDFs between peaks as a function of frequency separation. The color scale represents the maximum distance $D_{\max}$ from the K-S test, while the numerical values indicate the separation between peaks. A red star is assigned to any pair of peaks where $D_{\max} > 0.1$ and exceeds the 3σ noise level, marking the presence of a severe frequency drift between peak i and peak j . **Lower Right:** For comparison with the heatmap, the QQ plot of the frequency spectral distribution between peaks is presented, where the highlighted solid lines indicate the peak pairs exhibiting significant frequency drift just as the peak pairs marked in the lowerleft heatmap.

as

$$\text{PSNR} = \frac{I_{\text{peaks}}(t) - \bar{I}_{\text{noise}}}{\sigma_{\text{noise}}}, \quad (5)$$

where $I_{\text{peaks}}(t)$ represents the peak intensity, $\bar{I}_{\text{noise}}$ is the mean noise intensity, and σ_{noise} denotes the standard deviation of the noise. The point-mass lens model requires that $\text{PSNR}_i > \text{PSNR}_j$ for $i < j$. Similarly, taking FRB 20181028A in Fig. 3 as an example, peak 1 is significantly weaker than the other peaks, so it cannot form a point-mass lensed signal pair with any of the other peaks.

2.2.3. Frequency-Drift Test

Nevertheless, the presence of a spike in the ACF of the light curve does not guarantee a lensing origin, two peaks

with severe frequency drift are not indicative of gravitational lensing effect. Here we propose a Kolmogorov-Smirnov (K-S) test to verify whether severe frequency drift exists between the two peaks

$$D_{\max} = \sup_f [\text{CDF}(\bar{I}_i(f)), \text{CDF}(\bar{I}_j(f))] \quad (i \neq j), \quad (6)$$

where $\bar{I}_i(f)$ represents the normalized intensity profile of the i th peak in the frequency domain, which we treat as a random distribution in frequency, and CDF denotes its corresponding cumulative distribution function. At a significance level of $\alpha = 0.05$, given that the two peaks have equal sizes (we rebinned the 32 frequency channels of the CHIME data, resulting in $n_f = 512$ frequency channels for the sub-

sequent analysis), the critical value D_{crit} is approximately $D_{\text{crit}} = 1.36 \times \sqrt{2/n_f} \approx 0.1$. To further assess the resistance of D_{max} to outlier noise, we conducted a bootstrap analysis of the noise using $\mathcal{O}(10^3)$ iterations. Finally, a pair of peaks is characterized as exhibiting significant frequency drift if D_{max} satisfies both of the following conditions

$$\begin{cases} D_{\text{max}} > D_{\text{crit}}, \\ D_{\text{max}} > D_{n,\text{upp}}, \end{cases} \quad (7)$$

where $D_{n,\text{upp}}$ denotes the 3σ upper bound of the $D_{n,\text{max}}$ statistic obtained from K-S tests on the noise realizations. As illustrated in Fig. 3 for FRB 20181028A, significant frequency drift is present between peak 1 and peaks 35, while the peak pair (peak 3-4) corresponding to the distinct ACF spike exhibits no significant drift. This conclusion is further supported by the QQ plots, which reveal different patterns of drift among the drifting pairs—for instance, some show clear divergence at the head and tail quantiles, whereas others display an S-shaped deviation around the diagonal.

The detailed screening procedure is as follows: Firstly, we perform an ACF analysis on the light curve of each FRB dynamic spectrum. When a spike exceeding the 3σ threshold is identified, we search among the n_{peaks} brightest peaks in the light curve for a peak pair whose time delay matches the spike within an error tolerance of 2 ms. It is worth noting that if a spike is present but no corresponding peak pair can be found among these brightest n_{peaks} peaks, we exclude the lensing possibility. It is because the remaining peaks have very low SNR and are not considered meaningful for next analysis. If such peak pair is identified, we impose the following criteria: (i) the main peak must precede the secondary peak in time; (ii) the PSNR of secondary peak must be greater than 10 ($\text{PSNR}_2 > 10$) to ensure sufficient strength for subsequent analyses; and (iii) all the peak pairs must include the global maximum PSNR peak among all detected peaks. For any peak pair that satisfies these three initial criteria, we further examine whether significant frequency drift exists between the two peaks. Peak pairs that exhibit no significant frequency drift are tentatively regarded as lensed candidates for further investigation. FRB 20181028A is excluded under the above criteria because its ACF spike is dominated by peaks 2 and 3, and the matched pair lacks the global PSNR maximum as shown in Fig. 3. Based on the above criteria, i.e., ACF spikes, the arrival time order, and frequency drift, we initially screened 340 multi-peak FRBs for possible lensed candidates, identifying 11 candidates⁴. However, upon

reexamining these 11 FRBs, we found that in several cases the peaks were not clearly separated, appearing more like a single integrated signal with the peaks manifesting as spiky substructures (FRB 20221129B, FRB 20221216A), or exhibiting excessively diffuse frequency profiles for corresponding peak pair (FRB 20221129B). Consequently, under an almost rigorous initial screening, ultimately only 9 FRBs were selected for subsequent analysis: FRB 20190131D, FRB 20190915E, FRB 20200603B, FRB 20210117D, FRB 20210130C, FRB 20211115A, FRB 20220225C, FRB 20220424C, FRB 20230402B.

2.2.4. Hardness Test

The hardness test has been widely used to justify the lensing effect (Paynter et al. 2021; Veres et al. 2021; Wang et al. 2021; Yang et al. 2021; Lin et al. 2022), based on the hypothesis that the flux ratio between gravitationally lensed pulses should be frequency independent (Paczynski 1987). In this work, following the approach commonly adopted for lensed gamma-ray bursts (GRBs), we rebin the frequency channels into $k_{\text{HR}} + 2$ broad bands ($k_{\text{HR}} \in \{1, 2, 3, \dots\}$). When $k_{\text{HR}} = 1$, these bands are denoted by L, M, and H, corresponding to low, medium, and high frequencies, respectively. This rebinning procedure is designed for the hardness test: the frequency range of each FRB signal is determined from the RFI-free portion of the spectrum, i.e., $f \in [f_{\text{low}}, f_{\text{high}}]$, as provided in the CHIME/FRB Catalog 2. For each preliminary candidate, we first extract the timeresolved flux profiles in different frequency channels. Based on the autocorrelation analysis of the total burst profile, we split each frequencyresolved light curve into two similar episodes. We define a second episode of the same duration as the first one but shifted by a certain time delay. Subsequently, for each episode we define hardness ratios as

$$\text{HR}_{ij} = \frac{I_i - B_i}{I_j - B_j} \quad (j = i - 1, i \in [2, \dots, k_{\text{HR}} + 2]), \quad (8)$$

where I_i and B_i represent the total intensity (or flux) of the burst light curve and the background noise level for one episode, respectively. The uncertainty of HR_{ij} could be estimated considering Gaussian noise and the error propagation formula as follows

$$\sigma_{\text{HR}_{ij}} = \text{HR}_{ij} \sqrt{\frac{\sigma_{I_i}^2 + \sigma_{B_i}^2}{(I_i - B_i)^2} + \frac{\sigma_{I_j}^2 + \sigma_{B_j}^2}{(I_j - B_j)^2}}. \quad (9)$$

Here we require the preliminary candidate to pass the hardness test only when their HR_{ij} for different episodes are consistent with the mean value within the 1σ region. To determine the signal regions for the hardness test, we combined three times the full width at half maximum

⁴ All of the above and subsequent analysis procedures and results are provided at <https://github.com/Huan-Zhou-spec/MICRO-FRB>.

FRB Name	SNR	Δt (ms)	f range (MHz)	HR _{ML}		HR _{HM}		R_f	$M_{L,z}$ ($M_\odot$)	Rank S_{Lens}
				Episode 1	Episode 2	Episode 1	Episode 2			
FRB 20190131D	51.0	8.82	[400.2, 800.2]	0.99 $\pm$ 0.09	1.16 $\pm$ 0.26	0.63 $\pm$ 0.07	0.57 $\pm$ 0.15	2.57	466.50	2+4=6
FRB 20190915E	51.3	2.94	[521.0, 800.2]	3.51 $\pm$ 0.49	2.93 $\pm$ 0.53	<u>0.52 $\pm$ 0.05</u>	<u>0.65 $\pm$ 0.07</u>	✗	✗	✗
FRB 20200603B	57.4	9.80	[400.2, 582.2]	<u>0.60 $\pm$ 0.05</u>	<u>0.44 $\pm$ 0.05</u>	0.35 $\pm$ 0.07	0.35 $\pm$ 0.10	✗	✗	✗
FRB 20210117D	29.9	2.94	[414.5, 624.5]	<u>1.72 $\pm$ 0.20</u>	<u>0.78 $\pm$ 0.18</u>	0.35 $\pm$ 0.07	0.53 $\pm$ 0.21	✗	✗	✗
FRB 20210130C	37.7	2.94	[503.1, 800.2]	<u>2.31 $\pm$ 0.48</u>	<u>1.14 $\pm$ 0.16</u>	0.94 $\pm$ 0.14	0.77 $\pm$ 0.13	✗	✗	✗
FRB 20211115A	27.1	6.86	[400.2, 565.5]	1.47 $\pm$ 0.17	1.24 $\pm$ 0.24	0.37 $\pm$ 0.07	0.38 $\pm$ 0.12	1.77	609.45	2+1=3
FRB 20220225C	35.0	8.82	[400.2, 525.5]	<u>0.80 $\pm$ 0.10</u>	<u>1.29 $\pm$ 0.21</u>	0.48 $\pm$ 0.10	0.63 $\pm$ 0.11	✗	✗	✗
FRB 20220424C	36.2	6.86	[400.2, 800.2]	1.27 $\pm$ 0.14	1.10 $\pm$ 0.20	<u>0.43 $\pm$ 0.07</u>	<u>0.65 $\pm$ 0.13</u>	✗	✗	✗
FRB 20230402B	43.6	3.92	[400.2, 619.2]	<u>0.58 $\pm$ 0.03</u>	<u>0.26 $\pm$ 0.05</u>	0.29 $\pm$ 0.05	0.37 $\pm$ 0.17	✗	✗	✗

Table 1. Column 1: FRB name of TNS; Column 2: total SNR of FRB; Column 3: time delays between the matched peak pairs; Column 4: frequency range for hardness test; Column 5: HR_{ML} from two episodes; Column 6: HR_{HM} from two episodes; Column 7: optimal flux ratio; Column 8: lens redshifted mass; Column 9: microlensing candidate rank. Hardness test results for the 9 FRBs, a deviation exceeding the 1σ region is marked in underline. Only two candidates pass the hardness test $k_{\text{HR}} \geq 1$, i.e., FRB 20190131D and FRB 20211115A.

(FWHM) of the main peak with the time-delay positions. For FRB 20211115A, since a substructure (peak 2) is located between peak 1 and peak 3, we adopted only one FWHM instead of three. In the hardness test, we set the threshold at $k_{\text{HR},\text{min}} = 1$ (i.e., we require $k_{\text{HR}} > 1$) for the low, medium, and high-frequency bands. The final analysis results, presented in Table 1 and Fig. 4, yield two potential non-repeating FRBs that pass the hardness test, i.e., FRB 20190131D and FRB 20211115A. For both events, the maximum $k_{\text{HR},\text{max}}$ value reaches 2, which corresponds to the 4 frequency bands adopted in the hardness test.

2.3. Analysis of FRB Microlensing Candidates

2.3.1. FRB 20190131D

For FRB 20190131D, the dynamic spectrum is shown in the upper-left panel of Fig. 4 with the primary and secondary peaks labeled by green numbers. At the maximum value of $k_{\text{HR},\text{max}} = 2$, the HR_{ij} values in the two episodes are consistent within 1σ . In addition, the ACFs for the 4 frequency bands all exceed the 3σ threshold at the same time delay as the total light curve, yielding $k_{\text{ACF}} = 4$ shown in the upper-right panel of Fig. 4. Based on these results, we define a composite rank for microlensed FRB candidates as

$$S_{\text{Lens}} \equiv k_{\text{HR},\text{max}} + k_{\text{ACF}}, \quad (10)$$

which reflects the overall microlensing evidence and allows a direct comparison among candidates summarized in Table 1. If we relax the hardness test criterion by accepting any candidate whose HR_{ij} values are consistent within 2σ (For FRB 20190915E and FRB 20220424C, the HR_{HM} val-

ues are consistent within the 2σ range), then the maximum $k_{\text{HR},\text{max}}$ reaches 7 for FRB 20190131D. The signal can be divided into 9 frequency bins, while only 6 of them pass the ACF test at a significance level exceeding 3σ ($S_{\text{Lens}} = 7 + 6 = 13$). As the number of frequency bins increases, the SNR per bin inevitably decreases (see the total SNR values in Table 1), and beyond a certain division the noise contribution becomes dominant, which limits both the hardness test and the ACF test. It is noteworthy that Zhou et al. (2022a) also analyzed this source in detail but ultimately excluded it. Unlike our present approach which adopts a coarse four-bands division, their two-dimensional cross-correlation analysis divided the signal into 512 frequency channels, resulting in large errors and hence only weak correlation.

By adopting $\text{DM}_{\text{host}} + \text{DM}_{\text{src}} \in [0, 200]$ according to Eq. (1), the redshift range of the source is inferred to be $z_s \in [0.51, 0.67]$. The lens mass $M_{L,z}$ is estimated from the following expression based on the point mass model

$$M_{L,z} = \frac{\Delta t}{2 \left(\frac{R_f - 1}{\sqrt{R_f}} + \ln R_f \right)}, \quad (11)$$

where R_f is the amplification ratio between two images, which could be calculated with of the burst light curve and the background noise of two episodes ($R_f = (I_1 - B_1)/(I_2 - B_2)$). Given the lens redshift mass and the inferred redshift range, the lens mass range is estimated to be $M_L \in [280, 467] M_\odot$ falls in the IMBH regime.

2.3.2. FRB 20211115A

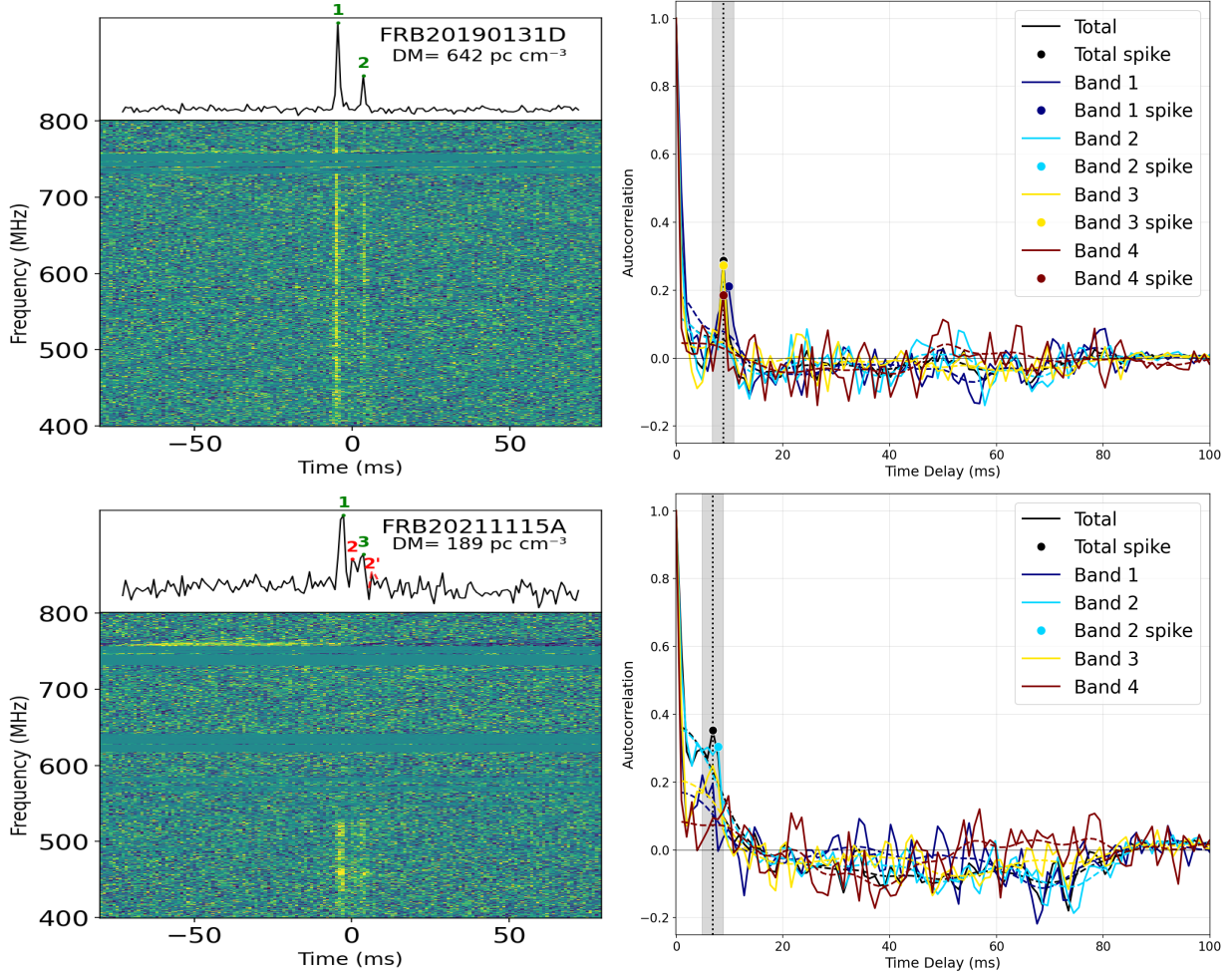


Figure 4. Similar to Upper panel of Fig. 3, but for FRB 20190131D, FRB 20211115A. In the lower-left dynamic spectrum for FRB 20211115A, peak 2 is shifted to the dashed peak based on the inferred flux ratio and time delay, and is labeled as 2'. The right panel shows the ACF with spikes within ± 2 ms range for the total and 4 frequency bands independently, where a colored dot on a spike indicates that the ACF exceeds the 3σ smooth threshold.

For FRB 20211115A, the situation presents different and more intriguing features. As shown in the dynamic spectrum in the lower-left panel of Fig. 4, peaks 1 and 3 form a primary-secondary peak pair that satisfies the lensing signal, while peak 2 can be regarded as a substructure of peak 1. Shifting peak 2 to its corresponding image position (peak 2') according to the flux ratio and time delay given in Table 1 places it at the red dashed line, where the light curve appears to exhibit a faint and ambiguous counterpart signal. However, owing to effect of noise, the spectral signature of the image corresponding to peak 2 is not clearly resolved in the two-dimensional dynamic spectrum. As shown in the lower right panel of Fig. 4, among the ACFs for the 4 frequency bands, only band 2 shows a spike at the same time delay as the total light curve above the 3σ threshold (while 3 spikes would be found if the threshold were reduced to 2σ). The microlensing candidate rank for this event is $S_{\text{Lens}} = 2 + 1 = 3$. Similar to FRB 20190131D,

if we relax the hardness test criterion by accepting candidates whose HR_{ij} values are consistent within 2σ , then for this source the maximum $k_{\text{HR,max}}$ reaches 4. The signal can be divided into 6 frequency bins, of which only 2 bands pass the ACF test at a significance level exceeding 3σ , giving a microlensing candidate rank $S_{\text{Lens}} = 4 + 2 = 6$. Therefore, compared with FRB 20190131D, FRB 20211115A not only shows a similar trend of decreasing test validity with finer band divisions, but also yields lower lensing significance, which may be explained by its relatively lower SNR. The redshift range of FRB 20211115A and lens mass ranges are obtained as $z_s \in [0, 0.13]$ and $M_L \in [539, 609] M_\odot$, respectively. Similarly to the case of FRB 20190131D, this lens mass again falls within the IMBH range.

3. CONSTRAINTS ON PBHS

The angular separation between different lensed images is of order $10^{-6}''$ for stellar-mass lenses, $1''$ for galaxy

lenses, and $1'$ for galaxy cluster lenses. The differences in path length and gravitational potential lead to time delays between these images of order 10^{-5} seconds for stellar-mass lenses, months for galaxy-scale lenses, and years for galaxy-cluster lenses. In [Muñoz et al. \(2016\)](#), it was first pointed out that the microlensing FRBs can be used to probe the compact dark matter, e.g. PBHs with masses as small as $\mathcal{O}(10 M_\odot)$. The PBH can be treated as a point mass with an Einstein radius θ_E . Although the spatial resolution in radio observations can reach a high level of $\sim 10^{-2}''$, it is still insufficient to distinguish split images for stellar-mass lenses. However, one can directly measure the time delay between lensed FRB signals. The formula for the time delay Δt is determined by the lens redshifted mass $M_{L,z} = (1 + z_L) M_{\text{PBH}}$, and the impact parameter $y = \beta/\theta_E$

$$\Delta t = 4M_{L,z} \left[\frac{y}{2} \sqrt{y^2 + 4} + \ln \left(\frac{\sqrt{y^2 + 4} + y}{\sqrt{y^2 + 4} - y} \right) \right]. \quad (12)$$

Δt must be larger than the width (w) of the observed signal. This requires y to be larger than a certain value $y_{\min}(M_{\text{PBH}}, z_L, w)$ according to Eq. (12). The lensing cross section due to a PBH lens is then given by:

$$\sigma(M_{\text{PBH}}, z_L, z_S, w) = \frac{4\pi M_{\text{PBH}} D_L D_{\text{LS}}}{D_S} \times [y_{\max}^2(R_{f,\max}) - y_{\min}^2(M_{\text{PBH}}, z_L, w)]. \quad (13)$$

The maximum value of the normalized impact parameter is found by requiring that the flux ratio of the two lensed images exceeds a reference value $R_{f,\max}$

$$y_{\max}(R_{f,\max}) = R_{f,\max}^{1/4} - R_{f,\max}^{-1/4}. \quad (14)$$

$R_{f,\max}$ should depend on the SNR of each observed FRB, which is reasonable since both SNR and flux ratio are crucial for identifying a lensed FRB event. The lensing optical depth due to a single PBH is:

$$\tau(M_{\text{PBH}}, f_{\text{PBH}}, z_S, w) = \frac{3}{2} f_{\text{PBH}} \Omega_{\text{DM},0} \int_0^{z_S} dz_L \frac{H_0^2}{H(z_L)} \times \frac{D_L D_{\text{LS}}}{D_S} (1 + z_L)^2 [y_{\max}^2(R_{f,\max}) - y_{\min}^2(M_{\text{PBH}}, z_L, w)], \quad (15)$$

where $H(z_L)$ is the Hubble function, f_{PBH} represents the fraction of PBHs in dark matter, and $\Omega_{\text{DM},0}$ is the present density parameter of dark matter. Given the statistically meaningful total FRB number N_{tot} , the expected number of lensed FRBs can be approximated by the sum of the lensing optical depths of all FRBs.

$$N_{\text{lensed FRB}} = \sum_{i=1}^{N_{\text{tot}}} \tau_i(M_{\text{PBH}}, f_{\text{PBH}}, z_{S,i}, w_i). \quad (16)$$

As shown in Figure 1, the width distribution and the SNR distribution ($R_{\text{max},f} = \text{SNR}/10$) together determine the

magnitude of the scattering cross section in Eq. (13). In addition, we find that the inferred redshifts of FRBs tend to be concentrated at low redshifts ($z < 2$) in Figure 1, which is one of the factors that significantly influences the results. If no genuine microlensed signal is found in the current data, the curve in the $(M_{\text{PBH}}, f_{\text{PBH}})$ parameter space that predicts at least three detectable lensed signals should be ruled out at the 95% confidence level. As shown in Fig 5, the mass can be tested down to $\sim 10 M_\odot$, and f_{PBH} is gradually constrained to 13% for the intermediate-mass PBH range $\gtrsim 300 M_\odot$ at the 95% confidence level by using SNR-dependent flux ratio thresholds. Owing to the increase in the number of CHIME/FRB sources from $\sim 5 \times 10^2$ to $\sim 4 \times 10^3$, our result constitutes an improvement of nearly an order of magnitude over previous constraints derived from CHIME/FRB catalog 1 ([Zhou et al. 2022a](#); [Krochek & Kovetz 2022](#)). Although current constraints are relatively weak, especially for small masses, much tighter constraints will be obtained from a large number of high-SNR FRBs in the near future ([Muñoz et al. 2016](#); [Laha 2020](#); [Connor & Ravi 2023](#)). Assuming both FRB 20190131D and FRB 20211115A originate from intermediate-mass PBH lensing system, a simple estimate of the PBH abundance in the relevant mass ranges can be made. We find that the PBH abundances in the mass ranges $\sim [280, 467] M_\odot$ and $\sim [539, 609] M_\odot$ are both constrained to approximately 4%, which are in significant tension with other observational constraints, e.g., effect of accretion (Accretion) and dynamical effects (Dynamical) as shown in Fig. 5.

4. CONCLUSIONS AND DISCUSSIONS

In this paper, we develop a pipeline to identify microlensed FRBs based on their dynamic spectra and apply it to FRB data from the CHIME/FRB Catalog 2, searching for microlensing signatures. The tests reveal evidence of correlation between the peaks of these FRBs. Intriguingly, we have reported two potential microlensing candidates, i.e. FRB 20190131D and FRB 20211115A. The inferred lens masses for these two signatures are $\sim [280, 467] M_\odot$ and $\sim [539, 609] M_\odot$, respectively. Both of them are in the range of IMBH. So far, there is no literature reporting the presence of galaxies or galaxy clusters in their line of sight directions of these two candidates. It indicates that the two IMBHs might be isolated and of primordial origins. On the basis of the primordial origin, we obtained a preliminary constraint on the PBH dark matter fraction: 1) these PBHs would constitute $\sim 4\%$ of the dark matter in the mass ranges $\sim [280, 467] M_\odot$ and $\sim [539, 609] M_\odot$; 2) if these candidates are not true gravitational lensing signals, the abundance of PBHs with masses $> 300 M_\odot$ could be constrained to $\sim 13\%$ at the 95% confidence level.

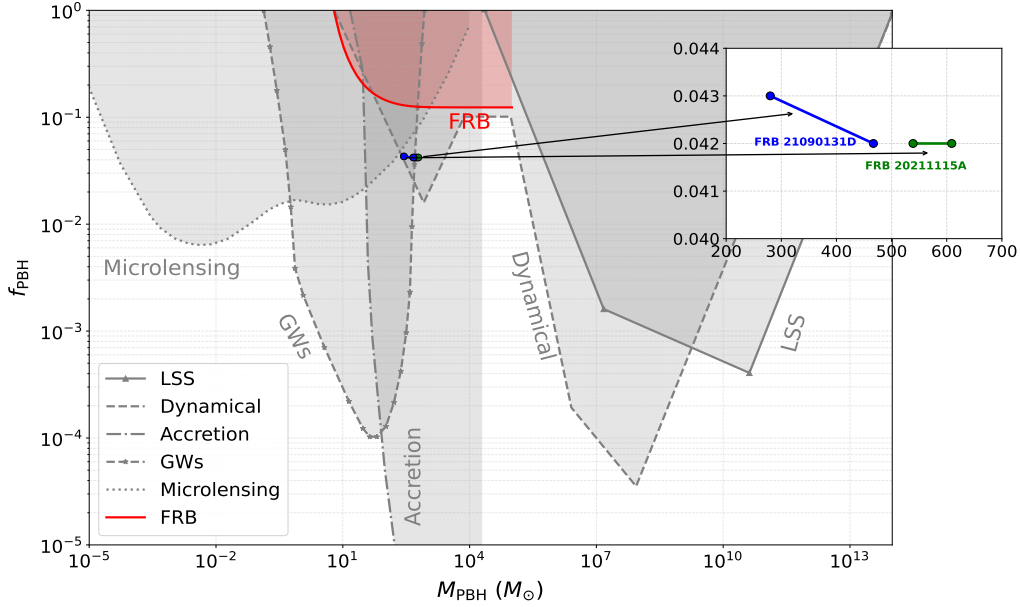


Figure 5. The upper limit on the PBH fraction f_{PBH} at the 95% confidence level as a function of PBH mass M_{PBH} in the range $[1, 10^5] M_{\odot}$, derived from the current CHIME/FRB catalog 2 under the assumption of no lensing FRBs. The subplot presents an approximate constraint on the PBH parameters under the hypothesis that FRB 20190131D and FRB 20211115A are lensed signals produced by PBHs. Other constraints are compiled from existing review Carr et al. (2026): these include limits from microlensing surveys (Microlensing), merger rate of binary black holes (GWs), effect of accretion (Accretion), dynamical effects (Dynamical), and the imprint of PBHs on large-scale structure (LSS).

Notably, Paynter et al. (2021) identified a milli-lensing signal in the light curve of GRB 950830 and inferred a lens mass of $\sim 10^4 M_{\odot}$, which lies within the IMBH range. Meanwhile, the GW event GW231123_135430 has been analyzed as a possible binary black hole lensed by a $\sim 600 M_{\odot}$ lens (Abac et al. 2025b). Together with the two possible IMBH candidates in our work, and given that FRB 20190320B was confirmed as a microlensed candidate with lens mass $\sim 400 M_{\odot}$ by analyzing hightimeresolution channelized data (Xiong et al. 2026), the number of evidence for the existence of IMBHs is constantly increasing. Alternative interpretations should also be considered. The observed subbursts could arise from intrinsic emission processes, such as doublepeaked profiles produced by magnetospheric activity or propagation effects within the FRB environment. Furthermore, as noted by Cordes et al. (2017), plasma structures within FRB host galaxies can also refract and lensing radio signals, giving rise to time delays and interference patterns analogous to gravitational lensing, thus offering a propagation-based explanation for FRB substructures and repetitions. Therefore, in future studies, it is essential to consider not only more complex lensing systems but also additional physical mechanisms that can produce lensing-like signals. Robust models of FRB light curves and spectral functions, incorporating Bayesian analyses and spectral index analyses analogous to

those in searching for lensed GRBs (Fast-Rise Exponential-Decay pulse model consistent with GRB fireball internal shocks (Paynter et al. 2021; Wang et al. 2021; Lin et al. 2022; Yang et al. 2021; Veres et al. 2021)), would provide additional evidence for validating microlensed FRBs. Complementary observations, such as polarization measurements or higher time resolution (a capability that suffers from a fundamental limitation of CHIME’s strategy for surveying FRBs, i.e., the sparse sampling around the peak of the burst), will be crucial for distinguishing genuine lensing events from false positives and for robustly validating microlensed FRBs.

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